

AgriFore: Data-Driven Agricultural Market and Yield Modeling for Kamareddy District, Telangana

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Abstract—AgriFore serves as a data-centered agricultural prediction system which focuses on creating crop yield and market price models for Telangana while providing district-level yield data for Kamareddy. Farmers and policymakers face a major challenge in the agricultural sector because they struggle with unpredictable price changes which become worse during times of extreme weather and when seasonal supply-demand patterns change and when market systems operate without coordination. The project integrates various types of heterogeneous agricultural data consisting of horticulture production records, daily weather observations, and Agricultural Market Committee (AMC) transaction data through a DuckDB-based ETL pipeline to support forecasting and market analytics. The system includes two principal modeling directions. The first is a Weather-to-Yield model, implemented as an artifact-backed district-specific XGBoost regressor for Kamareddy, which estimates `yield_per_acre_proxy` from seasonal rainfall, humidity, and weather-derived anomaly features, achieving a test R^2 of approximately 0.77 on 28 Kamareddy test records. The second, and primary served component, is a Market Price XGBoost base model that predicts monthly modal prices using lag features, rolling statistics, weather indicators, seasonal encodings, and market arrivals, achieving a test R^2 of 0.885 on a 2024 holdout of 4,264 eligible records. The model repository contains two offline validation branches, RidgeCV residual-correction layer that explores whether yield-linked signals improve price estimates and an LSTM autoencoder used for anomaly scoring on market sequences. A FastAPI backend and Next.js dashboard provide a locally runnable full-stack interface for market exploration, descriptive analysis, and base-model price estimation. Overall, AgriFore demonstrates how structured data engineering, gradient-boosted regression, and modular system design can be combined to support agricultural decision analysis while clearly separating deployed inference paths from offline experimental components.

Index Terms—Agricultural forecasting, XGBoost, LSTM autoencoder, crop yield prediction, price forecasting, DuckDB, weather-yield modeling, Kamareddy, Telangana agriculture, machine learning, time series analysis, anomaly scoring.

I. INTRODUCTION

Agriculture serves as a vital sector for India's economy, especially in Telangana where most of its residents rely on agriculture for their economic survival. The Kamareddy district which produces paddy, cotton, maize, and tomato crops functions as an agricultural system in which farmers encounter constant challenges related to their crop production and market value of their products. The market forecasting

process becomes difficult because weather patterns, seasonal supply-demand patterns, and market system deficiencies create unpredictable elements.

Traditional forecasting approaches such as ARIMA and Prophet have been widely used for time-series prediction; however, their ability to model nonlinear relationships and to combine weather data with crop production data and market arrival data remains restricted. The advanced capabilities of modern machine learning methods permit the creation of complex models that describe how various factors interact, yet current systems still treat yield prediction and price forecasting as distinct tasks that disregard their core causal relationship.

The existing challenges of the system require a solution that this paper introduces through AgriFore, an agricultural forecasting system that uses artificial intelligence to analyze data from multiple sources, and its two modeling components forecast both crop yields and market trends. The system first uses DuckDB-based ETL pipeline functionality to process horticulture, weather, and market datasets, and then analyzes them through two main modeling elements: an XGBoost model that predicts Weather-to-Yield and a Market Price XGBoost model. The system implements additional components that include an LSTM Autoencoder system for anomaly detection and a RidgeCV-based residual correction layer, which adjusts prices based on yield information.

The research delivers a major accomplishment through its development of an all-encompassing system that empowers users to create forecasts using visual and analytical resources. The proposed system architecture, as described in Section IV, shows how data moves from initial raw sources through the ETL process and modeling tiers until it reaches deployment through a FastAPI backend and Next.js frontend. The dashboard provides users with three functions that include displaying multi-commodity price trends, examining arrival-price relationships, working with prediction results which connect machine learning models to actual applications.

Researchers can use the system's visual and analytical tools to verify their research outcomes. The system demonstrates its capacity to handle multiple commodity datasets through its time-series price trend visualizations which display the data. Feature importance analysis provides a grouped artifact-level view of the served price model without implying causal

interpretation. The paper presents three main contributions that include a dual-model system that combines weather-based yield estimation with market price prediction, the creation of a DuckDB ETL system that processes different types of agricultural data, the development of a complete operational system that uses FastAPI for backend support and Next.js for user interface display and user interaction. The research verified its findings through three methods which included price trend examination, arrival-price relationship study, and feature importance assessment through grouped analysis.

- The system enables researchers to validate their findings by using its visual and analytical tools. The system demonstrates its capability to work with multiple commodity data sets through time-series price trend visualizations because it can display those data sets. Feature importance analysis provides a grouped artifact-level view of the served price model without implying causal interpretation.
- The development of a dual-model architecture that integrates weather-driven yield prediction with market price forecasting.
- The design of a DuckDB-based ETL pipeline for processing heterogeneous agricultural datasets.
- The implementation of an end-to-end deployed system with a FastAPI backend and Next.js dashboard for visualization and interaction.
- Empirical validation through price trend analysis, arrival-price correlation, and grouped feature-importance analysis.

II. RELATED WORK

A. Statistical Time Series Models

The researchers have employed conventional statistical time-series techniques to forecast market prices in the agricultural sector. The ARIMA model and its seasonal variant SARIMA remain popular baselines as they explain results through their well-established scientific foundations. Hyndman and Athanasopoulos [3] present a complete study of forecasting methods because their research shows that ARIMA models can accurately analyze stationary linear time series but fail to handle the complex nonlinear multiple factor patterns found in agricultural market data. Exponential smoothing methods [8] maintain their ability to assign different weights to present data but they share the same limitations which affect all other available techniques.

Taylor and Letham [4] created Prophet as a decomposable time series model which can manage seasonal patterns and holiday periods and trend change points for large scale applications. The commercial forecasting applications which use Prophet face limitations because its additive decomposition system fails to model the complicated relationships between weather patterns and market supply-demand patterns and regional agricultural trends which determine crop prices in Kamareddy districts.

B. Machine Learning Approaches

Chen and Guestrin [1] created XGBoost which serves as a tree boosting system that can scale to do structured data prediction tasks. XGBoost works effectively for agricultural forecasting because it can process different types of features handle missing data and model complex relationships between weather data and market indicators and crop type and district information.

Prokhorenkova et al. [2] developed CatBoost, which serves as an unbiased boosting algorithm that specifically handles datasets containing categorical features. The ordered boosting system of CatBoost together with its ability to process categorical data natively, creates a system that helps prevent overfitting while enhancing model performance, which establishes its value as a complementary system to XGBoost in ensemble setups.

Recent agricultural studies have implemented gradient boosting together with related ML methods to predict yield and price outcomes. Wang et al. [19] demonstrate that XGBoost-based modeling can be effective for agricultural yield classification under noisy field conditions while Liu et al. [20] show that machine-learning pipelines can capture nonlinear crop-growth and yield relationships more effectively than purely linear baselines. Sun et al. [21] review modern agricultural price forecasting methods for market-side forecasting while showing how tree ensembles and hybrid ML systems have become important tools for commodity-price analysis.

C. Deep Learning for Time Series

Recurrent neural networks, particularly LSTM architectures, have shown strong performance in sequential learning tasks [9]. The gating mechanisms of LSTM networks enable them to track long-term dependencies in price series, which makes them effective for modeling both temporal trends and seasonal patterns in commodity prices. Agricultural datasets contain noisy sequences that LSTM models cannot handle because they need large amounts of training data and their performance depends on accurate data.

Autoencoder architectures function as tools for detecting anomalies in financial markets and time series financial data. LSTM Autoencoders learn to reconstruct normal sequences, and instances with high reconstruction error are flagged as anomalies. The approach works effectively in agricultural markets because the markets experience sudden price movements which can result in two outcomes: data quality problems and market manipulation and authentic supply disruptions.

D. Hybrid and Ensemble Approaches

The recent research of this study investigates hybrid architectural designs which integrate deep learning systems with gradient boosting techniques. Athey and Imbens [9] present machine learning techniques which economists and market analysts can use to forecast economic conditions and market trends while showing how ensemble methods benefit from their various model components. The combination of LSTM temporal models and XGBoost feature models has proven to

enhance prediction accuracy beyond what each method can achieve independently [10], [11].

Recent studies on predicting agricultural prices have built on this work by using boosting-style ensembles that are specific to commodity data. Zhang et al. [22] discuss a boosting-ensemble method for predicting the prices of fresh agricultural products. This shows how important structured-data ensemble methods are to the market-price part of AgriFore.

E. Research Gap

Even with these improvements, there are still some gaps: (1) most current systems only look at price prediction on its own, without linking it to yield estimation, even though changes in yield due to weather are a big reason why market prices change; (2) few systems deal with the specific problems of Indian agricultural markets at the district level, where data quality and variety are big issues; (3) anomaly detection is rarely used in agricultural forecasting pipelines; and (4) end-to-end deployment architectures that bring ML predictions to non-technical stakeholders are still being worked on. AgriFore partially addresses these gaps in the current repository through a multi-model architecture, a functioning SQL/data layer, and a local full-stack application, with its deployed inference path currently focused on the base price model.

III. METHODOLOGY

A. Data Collection and Preprocessing

To make a single forecasting dataset for the Kamareddy district and the larger Telangana region, AgriFore combines data from three main sources:

1) Horticulture Production Data: The Telangana state government's Department of Horticulture provides Mandal-wise crop area, production and yield data for Telangana which covers the period from 2019-20 to 2022-23. The datasets include records that extend to every mandal located in various districts and contain information about horticulture crops which includes their cultivated area and total production and yield measurements. The system ingests raw data into DuckDB which then enables processing of data to achieve standardized district name matching and seasonal label normalization and calculation of yield-per-acre proxy values.

2) Weather Data: The system processes multiple years of daily rainfall and humidity records through cleaning, deduplication and aggregation operations. The system generates features which include total rainfall and average humidity and indicators of weather anomalies to show seasonal weather patterns.

3) Market Transaction Data: The Agricultural Market Committee (AMC) transaction records which include arrival volumes and modal prices aggregate into monthly summaries which researchers use to conduct price analysis across various commodities and markets.

The data pipeline uses DuckDB for its implementation which includes multiple SQL transformation stages that process data through the ingestion process and perform feature

engineering and outlier detection and create datasets. The process results in a structured analytical dataset which researchers use for downstream modeling.

B. Model 1: Weather-to-Yield Prediction

The first model in AgriFore's architecture establishes the link between seasonal weather patterns and agricultural crop production. The district-specific offline artifact is modeled as

$$y_i^{(\text{yield})} = f_1(\mathbf{w}_i, \mathbf{c}_i) + \epsilon_i \quad (1)$$

where $y_i^{(\text{yield})}$ is the `yield_per_acre_proxy` target, $\mathbf{w}_i$ denotes engineered seasonal weather features, and $\mathbf{c}_i$ denotes categorical identifiers such as district, crop, and season. The saved model is artifact-backed and offline rather than part of the served inference path.

The feature representation includes total rainfall, rainy-day counts, humidity levels, rainfall intensity, and rainfall-anomaly indicators. These weather conditions indicators provide both actual weather measurements and seasonal weather pattern deviations. The Kamareddy artifact uses a harvest-year holdout mechanism to prevent temporal leakage.

$$\mathcal{D}_{\text{train}}^{(1)} = \{i : \text{harvest_year}_i \in \{2020-21, 2021-22\}\}, |\mathcal{D}_{\text{train}}^{(1)}| = 61 \quad (2)$$

$$\mathcal{D}_{\text{test}}^{(1)} = \{i : \text{harvest_year}_i = 2022-23\}, |\mathcal{D}_{\text{test}}^{(1)}| = 28 \quad (3)$$

This strict temporal split keeps all test observations chronologically after the training window.

C. Model 2: Market Price Prediction (Base)

The second model focuses on monthly market-price regression using XGBoost. For observation i , the primary served component is written as

$$y_i^{(\text{price})} = f_2(\mathbf{x}_i, \mathbf{c}_i) + \epsilon_i \quad (4)$$

where $y_i^{(\text{price})}$ denotes the monthly modal price, $\mathbf{x}_i$ contains numeric covariates, and $\mathbf{c}_i$ contains district, AMC, crop, and season identifiers.

Key features include lagged prices, rolling averages, monthly rainfall, humidity indicators, arrival volumes, and cyclical month encodings. The temporal features are constructed per market grouping g :

$$\text{month_sin} = \sin\left(\frac{2\pi m}{12}\right), \quad \text{month_cos} = \cos\left(\frac{2\pi m}{12}\right) \quad (5)$$

$$\text{lag}_k(t, g) = y_{t-k}^{(\text{price})}(g) \quad (6)$$

$$\text{roll_mean}_w(t, g) = \frac{1}{w} \sum_{j=1}^w y_{t-j}^{(\text{price})}(g) \quad (7)$$

where m is calendar month, $k \in \{1, 2, 3, 6, 12\}$, and $w \in \{3, 6\}$. These definitions make the history-based feature construction explicit without implying a purely autoregressive model.

Both regression models share the same XGBoost backbone. After preprocessing, the prediction is an additive tree ensemble:

$$\hat{y}_i = \sum_{k=1}^K f_k(\mathbf{x}_i), \quad f_k \in \mathcal{F} \quad (8)$$

At boosting round t , the regularized objective is

$$\mathcal{L}^{(t)} = \sum_{i=1}^n \ell(y_i, \hat{y}_i^{(t-1)} + f_t(\mathbf{x}_i)) + \Omega(f_t) \quad (9)$$

with

$$\Omega(f) = \gamma L + \frac{1}{2} \lambda \sum_{j=1}^L w_j^2 \quad (10)$$

where L is the number of leaves and w_j is the score of leaf j . Using the standard second-order approximation,

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^n \left[g_i f_t(\mathbf{x}_i) + \frac{1}{2} h_i f_t(\mathbf{x}_i)^2 \right] + \Omega(f_t) + \text{const} \quad (11)$$

and the optimal weight for leaf j is

$$w_j^* = - \frac{\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda} \quad (12)$$

In this concise form, we have the regularized boosting formulation in a few statements, without working through the usual appendix-style procedure.

The model requires arrival volume data to function as its supply-side component. The authors of this study present arrival data together with price data which they analyze through a static model that shows different results for various commodities and locations and time periods.

The deployed evaluation uses a time-based holdout over eligible monthly rows:

$$\mathcal{D}_{\text{train}}^{(2)} = \{i : \text{month_start}_i \leq 2023-12-01 \wedge \text{eligible}(i)\}, |\mathcal{D}_{\text{train}}^{(2)}| = 28,571 \quad (13)$$

$$\mathcal{D}_{\text{test}}^{(2)} = \{i : \text{month_start}_i \geq 2024-01-01 \wedge \text{eligible}(i)\}, |\mathcal{D}_{\text{test}}^{(2)}| = 4,264 \quad (14)$$

where $\text{eligible}(i)$ denotes rows with non-null `season_norm` in the modelable Kharif/Rabi window, excluding the April/May rows that fall outside this season mapping. Validation for the offline adjustment experiment further uses forward temporal blocks to generate leakage-free out-of-fold residuals.

D. Model 2: Yield-Adjusted Price Prediction

A secondary residual-correction layer is implemented as an offline experiment to refine base-model predictions. Given base prediction $\hat{y}_i^{(\text{base})}$ and observed price y_i , the residual target is

$$r_i = y_i - \hat{y}_i^{(\text{base})} \quad (15)$$

The adjustment model fits a Ridge regression on auxiliary features $\mathbf{z}_i$:

$$\hat{r}_i = \mathbf{z}_i^\top \beta \quad (16)$$

with coefficients obtained from

$$\beta^* = \arg \min_{\beta} \left\{ \sum_{i=1}^{n_{\text{adj}}} (r_i - \mathbf{z}_i^\top \beta)^2 + \alpha \|\beta\|_2^2 \right\} \quad (17)$$

leading to the adjusted offline estimate

$$\hat{y}_i^{(\text{final})} = \hat{y}_i^{(\text{base})} + \hat{r}_i \quad (18)$$

The adjustment layer uses out-of-fold predictions for its training process which prevents data leakage. The system functions as an offline testing module but its performance metrics fail to demonstrate any progress beyond the base model which is currently active.

E. LSTM Autoencoder for Anomaly Scoring

An LSTM autoencoder is implemented as an offline validation component for scoring unusual price sequences. For an input window $\mathbf{X}_j \in \mathbb{R}^{T \times d}$ and its reconstruction $\hat{\mathbf{X}}_j$, a simple reconstruction score is

$$e_j = \frac{1}{Td} \sum_{t=1}^T \sum_{f=1}^d (X_{j,t,f} - \hat{X}_{j,t,f})^2 \quad (19)$$

Higher values indicate poorer reconstruction in the sampled sequence. The deployed system does not include this component because it exists as an offline anomaly-scoring artifact which has not been tested for production use.

F. Evaluation Metrics

Model performance is evaluated using standard regression metrics:

- **Root Mean Squared Error (RMSE):** It quantifies the magnitude of the average errors in predictions while penalizing greater errors more severely.
- **Mean Absolute Error (MAE):** It can be possible to set the quality of the absolute difference from the predicted value to the actual value.
- **R^2 Score:** Representing the amount of the variance that the model explains, hinting at the performance of some sort of universal predictive act.

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad \text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (20)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (21)$$

IV. SYSTEM ARCHITECTURE

AgriFore implements its system design through a three-tier modular architecture which includes three different operational components: a data pipeline layer, a backend API layer and a frontend presentation layer. The design achieves three main benefits through its implementation which allows for different operational processes to function independently while making it easier to perform maintenance tasks. Figure 1 shows the served inference boundary which the repository displayed on

1 April 2026. The document identifies the M2 base-model runtime path as the main operational route while stating that offline experimental modules exist as separate entities.

A. Data Pipeline — DuckDB ETL

The data pipeline uses DuckDB which functions as an analytical database engine that provides high-performance processing for complex structured data queries. The embedded system of DuckDB enables users to access its columnar storage and vectorized query execution features without needing a database server.

The current runnable pipelines use 17 active SQL scripts organized in sequential phases, with one superseded weather-loader SQL file retained for reference:

- **Phase 1 (Scripts 00–10):** The process begins with the raw data ingestion from horticulture CSV files which leads to the development of a view that allows for production and area extraction using standardized district names and seasonal identifiers.
- **Phase 2 (Scripts 15–18):** The process of weather data loading and cleaning deduplication work, along with weather data processing, occurs at three levels of daily and monthly and seasonal time measurements. The district name matching diagnostics process verifies that production data and weather datasets maintain consistent matching results.
- **Phase 3 (Scripts 20–40):** The data quality assessment process includes three main activities which involve conducting crop coverage analysis and stability profiling and using MAD-based outlier detection methods together with robust z-scores and seasonal alignment procedures for harvest-year rollups.
- **Phase 4 (Scripts 50–66):** The project contains five data components which include the Model 1 weather-yield dataset and market data and monthly data and seasonal calendar data and the Model 2 base and adjustment datasets.

The resulting database contains over 30 tables and views including `t_model1_dataset`, `v_model2_base_dataset`, `v_model2_adjust_dataset`, `t_market_monthly`, and `t_weather_district_season`; the checked-in DuckDB file is approximately 123 MB.

B. Backend - FastAPI Server

The backend system uses FastAPI which is a contemporary Python web framework that generates OpenAPI documentation and uses Pydantic for type validation and supports asynchronous request processing. The existing API provides several endpoints which include a health check and functional endpoints that are organized into different groups.

Reference Data Endpoints:

- `GET /commodities` - Lists all commodities with category classification
- `GET /markets` - Lists AMC market-yard entries with districts

- `GET /districts` - Lists Telangana districts

Dashboard Endpoints:

- `GET /market-overview` - Aggregated statistics
- `GET /top-commodities` - Top commodities by volume
- `GET /market-clusters` - Regional market overview

Chart Data Endpoints:

- `GET /price-trends` - Multi-commodity price trends
- `GET /seasonal-data` - Monthly seasonal analysis
- `GET /arrival-data` - Arrival vs price correlation

Prediction Endpoints:

- `POST /predict-price` - Price prediction with fallback mechanisms
- `POST /ask-agent` - Agricultural Q&A endpoint
- `GET /health` - System health check

The server connects to DuckDB in read-only mode and loads the base XGBoost price model at startup. The residual correction and anomaly-scoring modules are not currently part of deployed inference.

C. Frontend — Next.js Dashboard

The frontend is built using Next.js with TypeScript and Tailwind CSS, providing an interactive dashboard for visualization and analysis.

The dashboard consists of three main pages:

Dashboard Page: The system presents summary statistics together with price trends and information about top commodities and regional clusters. The repository shows repo-backed monthly market history data for crops through its active monthly market history data which demonstrates both market behavior and price changes.

Analysis Page: The system delivers two types of analysis which include seasonal analysis together with arrival-price relationship visualizations. The connection between arrivals and price data in the plotted sample gets demonstrated through Figure 3 however this finding should not be seen as proof of an existence of a universal inverse relationship.

Prediction Page: Allows users to input parameters and receive AI-driven price predictions.

The frontend communicates with the backend via API calls, enabling real-time data visualization and prediction.

D. System Workflow

The system workflow operates as follows:

- 1) Data ingestion from horticulture, weather, and market datasets
- 2) ETL processing using DuckDB SQL pipelines
- 3) Model training using prepared datasets
- 4) Model serving through FastAPI
- 5) API-based data delivery to frontend
- 6) User interaction through the dashboard

This architecture enables seamless data flow from raw data sources to actionable insights while maintaining modularity and scalability.

AgriFore fresh architecture alignment

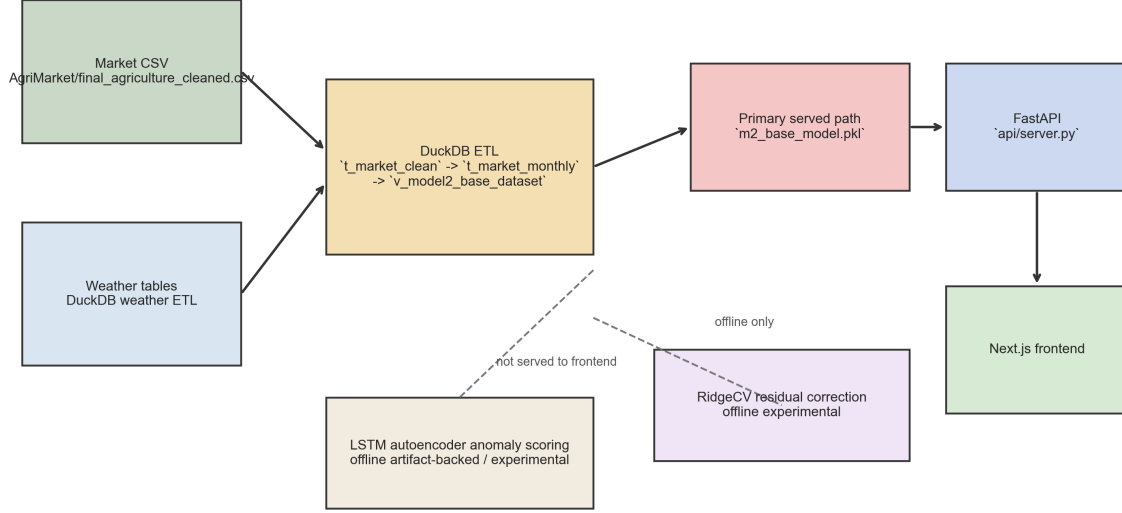


Fig. 1. Served-path architecture of the AgriFore repository as reproduced on 1 April 2026, separating the primary M2 base-model inference path from offline experimental modules.

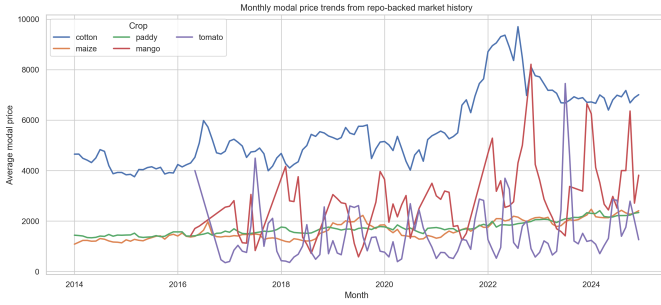


Fig. 2. Descriptive monthly modal price trends for representative crops from repo-backed market history; coverage windows may differ across crops.

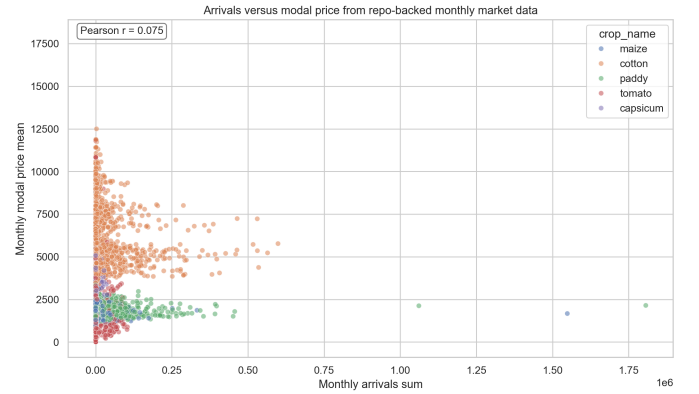


Fig. 3. Descriptive relationship between monthly arrivals and modal price in repo-backed market data for selected crops; the annotated Pearson correlation summarizes association in the plotted sample only.

V. RESULT ANALYSIS

The AgriFore forecasting system performance evaluation needed extensive testing because researchers worked with multiple years of agricultural datasets from Telangana government sources and IMD weather records. The researchers assessed the models through standard regression metrics which they applied to test sets created with different time periods to replicate actual forecasting conditions.

A. Evaluation Metrics

Three standard regression metrics were used to assess forecasting performance:

- **Root Mean Squared Error (RMSE):** Measures average magnitude of error, with lower values indicating better accuracy. RMSE penalizes larger errors more heavily than MAE.

- **Mean Absolute Error (MAE):** Reflects the average absolute prediction error, providing an interpretable measure of typical forecast deviation.
- **R² Score:** Represents the proportion of variance explained by the model, with a score closer to 1.0 indicating better goodness-of-fit.

B. Model 1 Results: Weather-to-Yield Prediction (Kamareddy District)

Model 1 was evaluated on the Kamareddy district with training on 61 records from years 2020–21 and 2021–22, and testing on 28 records from year 2022–23. Table I presents the results.

The near-perfect training R² (0.9999) with a substantially lower test R² (0.77) suggests some degree of overfitting,

TABLE I
MODEL 1 WEATHER-TO-YIELD PERFORMANCE FOR KAMAREDDY
DISTRICT (ARTIFACT-BACKED)

Metric	Training Set	Test Set
R ² Score	0.9999	0.7702
RMSE	0.0009	3.9370
MAE	0.0008	2.0582

TABLE II
MODEL 2 MARKET PRICE PREDICTION PERFORMANCE ON THE ELIGIBLE
2024 HOLDOUT

Metric	Training Set	Test Set
R ² Score	0.9574	0.8849
RMSE	549.02	1170.63
MAE	342.50	610.09

which is expected given the relatively small training dataset. However, the test R² indicates that the model explains approximately 77% of the variance in crop yields using weather features.

C. Model 2 Results: Market Price Prediction (XGBoost Base)

Model 2 was trained on 28,571 monthly records and tested on 4,264 records covering multiple commodities and markets. Table II presents the results.

The Model 2 base XGBoost achieves strong predictive performance, with a test R² of 0.885, indicating effective modeling of price dynamics across commodities.

Figure 4 displays the combined feature importance data which was obtained from the M2 base model artifact that had been deployed. The transformed features were returned to their original parent groups to create visualizations which serve only as descriptive artifacts and not as explanations of price movement.

D. Model Comparison

Table III compares the AgriFore models against traditional forecasting approaches.

Table III is included as literature context rather than as a direct benchmark. No ARIMA, Prophet, or standalone LSTM baselines were retrained on the AgriFore data in this manuscript.

E. LSTM Autoencoder Anomaly Scoring

The repository contains an offline LSTM autoencoder validator, which evaluates rare price patterns. The system has developed this component and completed its assessment, but it remains absent from the operational system.

F. Prediction Deployment

The deployed POST /predict-price endpoint implements a three-strategy fallback mechanism:

- 1) **Base Model Inference:** Real-time prediction using the trained XGBoost model
- 2) **Pre-computed Lookup:** Fallback to stored predictions
- 3) **Historical Average:** Final fallback using historical price averages

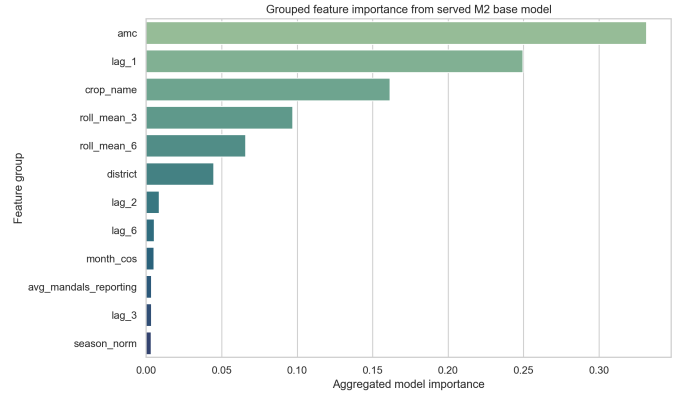


Fig. 4. Grouped feature-importance summary extracted from the served M2 base model artifact; transformed features were aggregated back to parent groups for readability and should not be interpreted causally.

TABLE III
PERFORMANCE CONTEXT FOR AGRIFORE AND LITERATURE-REPORTED
RANGES

Approach	Typical Range	R ²	Notes
ARIMA [3]	0.45–0.65		Linear; limited nonlinear capacity
Prophet [4]	0.55–0.70		Limited exogenous-variable support
Standalone LSTM	0.70–0.80		Data-hungry and noise-sensitive
Standalone XGBoost	0.75–0.85		Strong structured-data baseline without explicit sequence modeling
AgriFore Model 2 (XG-Boost + lag features)	0.885		Pooled eligible 2024 holdout result in this repository
AgriFore Model 1 (Weather→Yield, Kamareddy)	0.770		Small-sample district-specific artifact-backed result

This fallback strategy ensures robustness in real-world deployment scenarios. For a request q , the primary served estimate can be written schematically as $\hat{y}_q^{(1)} = f_2(\tilde{x}_q)$, where $\tilde{x}_q$ is constructed from the latest available historical market and weather context for the requested crop-market pair. The endpoint should therefore be interpreted as a history-based estimation service with fallback logic rather than as a strict year-conditioned future-forecast engine. If Strategy 1 fails, the remaining fallbacks can be summarized as

$$\hat{y}_q^{(2)} = \frac{1}{|M_q|} \sum_{j \in M_q} \hat{y}_j^{(2024\text{-csv})}, \quad \hat{y}_q^{(3)} = \frac{1}{|H_q|} \sum_{j \in H_q} y_j^{(\text{historical})} \quad (22)$$

where M_q matches the requested crop–district–market tuple in the stored 2024 predictions file and H_q matches the historical crop–district–market–month subset in DuckDB. These fallback outputs indicate provenance rather than calibrated uncertainty.

VI. CONCLUSION AND FUTURE WORK

A. Conclusion

We developed AgriFore as a multi-component repository which stores agricultural price and yield forecasting models that use Telangana data, including a Kamareddy yield artifact

which operates at the district level and a statewide market-price base model. The repository holds two components which include a Kamareddy Weather-to-Yield XGBoost artifact that achieved a test R^2 result of approximately 0.77 and a Market Price XGBoost base model which reached a test R^2 of 0.885. The repository holds two components which include an offline LSTM autoencoder validator and an offline RidgeCV residual-correction layer, but these components have not been added to the backend inference system which is currently operational.

The DuckDB ETL stack uses 17 current SQL stage scripts to process horticulture records and weather observations and market transaction logs, while the repository keeps one weather-loader script that is no longer active. The FastAPI backend with Next.js dashboard provides users with a functioning local application, which enables data access and market analysis and base-model price prediction. The repository maintains a conservative deployment claim, because the backend system currently operates only the base market model, while both adjustment and anomaly modules stay in offline or testing mode.

The reported base accuracy of Model 2 gets validated through evaluation artifacts which encompass 117 commodities across 169 Agricultural Market Committee entries. The repository establishes a complete base for structured data forecasting because it includes both a complete ETL system and an operational application framework while some elements for causal analysis and anomaly detection remain in testing.

B. Future Work

Future development of AgriFore will focus on the following enhancements:

- 1) **Real-Time Data Integration:** The system uses live APIs from AGMARKNET [5] and IMD [6] to collect weather and price data in near real-time which allows the system to update its models and produce forecasts in real time.
- 2) **Transformer-Based Architectures:** The research investigates Temporal Fusion Transformers (TFT) to create better temporal attention systems which function through automated feature selection that replaces the need for manual lag feature development.
- 3) **Explainability Tools:** Blending SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) for higher interpretability and user trust- so as to maintain the same things in different words.
- 4) **Geographic Expansion:** Expanding the Kamareddy-specific Model 1 workflow with other district-level datasets contained in the repository.
- 5) **Mobile Application Development:** Extending the Next.js dashboard to mobile platforms using React Native, providing accessibility to farmers deep into rural areas.
- 6) **Supply Chain Integration:** Incorporating downstream forecasting for procurement timing and logistics optimization based on predicted price and yield trends.

- 7) **Continuous Learning and Uncertainty Modeling:** Transitioning the Model 2 base system toward incremental learning through XGBoost warm-start capabilities and implementing Conformal Prediction or Quantile Regression for automatic dynamic confidence intervals.

By continuing to refine and scale this system, and by evaluating offline components more rigorously before any deployment claim is extended, AgriFore has the potential to evolve into a more comprehensive decision-support system for agricultural stakeholders.

APPENDIX A WORKED FORECAST EXAMPLE FOR A SINGLE KAMAREDDY CROP

A. Setup

Appendix A presents one repo-backed, single-crop implementation example to show how the two supervised formulations map onto a concrete Kamareddy case without implying holdout validation for that local slice. The selected pair is Tomato in Kamareddy district, using Madnoor AMC for the market-side example.

TABLE IV
APPENDIX A SETUP FOR THE SINGLE-CROP KAMAREDDY EXAMPLE

Field	Value
Crop	Tomato
District	Kamareddy
AMC	Madnoor
Model 1 period	2022–23 Kharif
Model 2 period	July–August 2022
Example status	Single-crop Kamareddy implementation example

B. Model 1 Weather-to-Yield Worked Observation

The Model 1 portion remains artifact-backed. The repository contains a saved Kamareddy yield artifact, but it fails to provide users with a single path that enables them to reproduce each exact saved row-level prediction. The worked observation therefore presents a verified seasonal input row and its observed target:

$$\hat{y}_i^{(1)} = \hat{f}^{(1)}(\tilde{\mathbf{x}}_i^{(1)}) \quad (23)$$

TABLE V
APPENDIX A MODEL 1 WORKED OBSERVATION FOR TOMATO IN
KAMAREDDY

Quantity	Value
rain_total_mm	916.20
rainy_days_total	48
hum_min_avg	65.3989
hum_max_avg	94.7172
avg_mandals_reporting	22.0
days_covered	122
months_covered	4
rain_intensity	19.0875
humid_range	29.3183
Observed yield_per_acre_proxy	12.0000

The artifact metadata also lists `rain_anom_mm` and `rain_dev_pct` as model inputs, but those terms are not

stored directly as row values in the final dataset table. For that reason, the appendix treats the displayed row as a verified observation with an observed target rather than as a canonically reproducible per-row artifact prediction.

C. Model 2 Single-Sample Price Example

The market-side example uses the real Tomato–Kamareddy–Madnoor row for August 2022 and a direct local inference from the verified saved Model 2 artifact:

$$\hat{y}_i^{(2)} = \hat{f}^{(2)}(\tilde{\mathbf{x}}_i^{(2)}) \quad (24)$$

TABLE VI
APPENDIX A MODEL 2 KEY SUB-VECTOR FOR TOMATO AT MADNOOR
AMC

Quantity	Value
month_start	2022-08-01
district	Kamareddy
amc	Madnoor
crop_name	Tomato
Actual modal_price_mean	1500.0000
arrivals_sum	450.0
n_days	4
rain_sum_mm	110.00
rainy_days	11
hum_min_avg	71.3046
hum_max_avg	97.3534
avg_mandals_reporting	22.0
month_num	8
month_sin	-0.8660
month_cos	-0.5000
lag_1	1366.6667
lag_2, lag_3, lag_6, lag_12	missing
roll_mean_3, roll_mean_6	missing
Verified saved-model prediction	1743.4164

The repository contains only two available monthly rows for this crop–district–market pair which includes data from July and August 2022. The system uses model pipeline imputation path to manage several lag and rolling features which cannot be accessed locally.

D. Local Error Illustration

The local error note shows sparse results because only two Tomato–Madnoor rows exist, which makes the note function as a training-period example instead of a complete evaluation tool.

TABLE VII
APPENDIX A LOCAL ERROR ILLUSTRATION FOR TOMATO AT MADNOOR
AMC

Month	Actual Price	Predicted Price	Absolute Error
2022-07	1366.6667	3271.2236	1904.5570
2022-08	1500.0000	1743.4164	243.4164

$$\text{MAE}_{\text{local}} = \frac{|1366.6667 - 3271.2236| + |1500.0000 - 1743.4164|}{2} \quad (25)$$

$$= 1073.9867$$

$$\text{RMSE}_{\text{local}} = \sqrt{\frac{(1366.6667 - 3271.2236)^2 + (1500.0000 - 1743.4164)^2}{2}} \quad (26)$$

$$= 1357.6798$$

These local quantities are useful only as an appendix-style numerical illustration of how the saved model behaves on an extremely sparse crop–location slice. They are not a substitute for the pooled statewide holdout results in Table II.

E. Interpretation

Appendix A shows that the two supervised methods from the paper connect to a specific Kamareddy crop which uses repository-backed values and authentic local Model 2 inference. The study does not establish Tomato-specific holdout accuracy or causal connections or validation results observed in March 2026 for Kamareddy. The Model 1 row maintains its artifact-backed status because it fails to produce canonical per-row prediction results while the Model 2 local slice contains fewer elements compared to the statewide assessment that the main paper employed.

APPENDIX B FRESH MARCH 2026 EXTERNAL VALIDATION SNAPSHOT

TABLE VIII
APPENDIX B MATCHED CROP–MARKET PAIRS IN THE FRESH MARCH
2026 COMPARISON

Crop	District	Market	Prediction	Observed	Absolute Error
Cotton	adilabad	adilabad	7101.05	6268.33	832.72
Maize	siddipet	husnabad	2483.48	1950.00	533.48
Tomato	rangareddy	chevella	1803.47	1000.00	803.47
Capsicum	warangal urban	warangal	4653.25	4750.00	96.75
Cabbage	warangal urban	warangal	2063.03	1350.00	713.03
Brinjal	warangal urban	warangal	1963.75	1200.00	763.75

Appendix B summarizes the fresh 1 April 2026 comparison between newly generated March 2026 model outputs and sparse external March 2026 observations. The small matched set needed external observed pages because the checked-in DuckDB market history ended in December 2024 which made Repo-backed March 2026 market rows unavailable.

TABLE IX
APPENDIX B SUMMARY OF FRESH MARCH 2026 EXTERNAL
COMPARISON

Quantity	Value
Matched rows	6
Unmatched rows	2
MAE	623.8677
RMSE	673.8790
Absolute MAPE (%)	39.9149
R ²	0.778

The prediction pairs that were unmatched among the fresh predictions pairs were Paddy–Husnabad and Mango–Gaddiannaram. The appendix serves as an external validation note which does not establish a definite benchmark. The March 2026 pages showed external observations but had limited

coverage and several observations showed closer similarity to sampled March entries than to full-month transaction aggregates. The Kamareddy worked example in Appendix A should not include this concept because it lacks validation for Kamareddy-based observations during March 2026.

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